

NURBEK TASTAN

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google scholar: <https://scholar.google.com/citations?user=8eQlTUQAAAAJ>

PROFILE

Doctoral researcher with a strong foundation in machine learning, optimization, and trustworthy systems. My research focuses on **trustworthy and efficient machine learning**, with particular emphasis on **privacy**, **fairness**, **efficiency**, and **robustness** in collaborative settings. I also develop methods to make large-scale models, including **large language models**, more efficient (in pre-training, fine-tuning and inference) while also preserving utility and confidentiality. **Currently** expanding my research toward agentic AI and LLMs, with a focus on the trustworthiness and efficiency of such systems.

Passionate about building machine learning systems that are not only performant, but also **privacy-aware**, **equitable**, and **computationally viable** for real-world deployment.

EDUCATION

PhD in Machine Learning, Mohamed bin Zayed University of Artificial Intelligence 2023 - Present
Abu-Dhabi, United Arab Emirates

Relevant Coursework: Foundations and Advanced Topics in Machine Learning, Advanced Probabilistic and Statistical Inference, Advanced Topics in Continuous Optimization, Federated Learning, Safe and Robust Computer Vision, Advanced Machine Learning

MSc in Machine Learning, Mohamed bin Zayed University of Artificial Intelligence 2021 - 2023
Abu-Dhabi, United Arab Emirates

Relevant Coursework: Mathematical Foundations for AI, Machine Learning, Advanced ML, Trustworthy AI, Probabilistic and Statistical Inference, Causality, Reinforcement Learning, Optimization

BSc in Systems of Information Security, International Information Technology University 2017 - 2021
Almaty, Kazakhstan

Relevant Coursework: Programming (Python, C++, Java), ML, Cybersecurity, Math
[Summa cum laude] GPA **3.94**/4.00 (Top 1 of 2021 graduating class)

EXPERIENCE

Visiting Researcher Feb 2026 - Apr 2026
University of Cambridge *Cambridge, United Kingdom*

- Hosting lab: CaMLSys

Lead Teaching Assistant Jan 2022 - May 2025
MBZUAI *Abu-Dhabi, United Arab Emirates*

- Courses: [DS702: Big Data Processing], [MTH701: Mathematical Foundations of Artificial Intelligence]
- More information on <https://tnurbek.github.io/teaching/>.

Data Scientist May 2020 - Sep 2021
Healthcare IT *Almaty, Kazakhstan*

SKILLS

Languages	Python, C++, Scala, SQL (PostgreSQL, ClickHouse, Cassandra)
Frameworks	PyTorch, Tensorflow, Spark/PySpark, HuggingFace Transformers
Research Focus	Machine Learning, Trustworthy ML, Efficient ML, Optimization, Federated Learning, Large Language Models, Mixture-of-Experts, Foundation Models, Scalable ML, Efficient [Pre-Training, Fine-tuning, Inference], Multi-Agent Systems, Agentic AI
Additional Tools	Docker, Git, Linux, W&B, bash, Slurm, vLLM, DDP/FSDP/DeepSpeed

SELECTED PUBLICATIONS

MoSE: Mixture of Slimmable Experts for Efficient and Adaptive Language Models

Nurbek Tastan, Stefanos Laskaridis, Karthik Nandakumar, Samuel Horváth. Under Review.

Stochastic Self-Organization in Multi-Agent Systems

Nurbek Tastan, Samuel Horváth, Karthik Nandakumar. ICLR, 2026.

LoFT: Low-Rank Adaptation That Behaves Like Full Fine-Tuning

Nurbek Tastan, Stefanos Laskaridis, Martin Takáč, Karthik Nandakumar, Samuel Horváth. ICLR, 2026.

ICML 2025 Workshop, Tiny Titans: TTODLer-FM@ICML. **Best Paper Honorable Mention.**

MoLM: Mixture of LoRA Markers

Samar Fares, Nurbek Tastan, Noor Hussein, Karthik Nandakumar. ICLR, 2026.

CYCLE: Choosing Your Collaborators Wisely to Enhance Collaborative Fairness in Decentralized Learning.

Nurbek Tastan, Samuel Horváth, Karthik Nandakumar. TMLR, 08/2025.

Aequa: Provably Fair Federated Learning using Slimmable Networks.

Nurbek Tastan, Samuel Horváth, Karthik Nandakumar. ICML, 2025.

A Framework for Double-Blind Federated Adaptation of Foundation Models.

Nurbek Tastan, Karthik Nandakumar. ICCV, 2025. ICLR 2025 Workshop (MCDC@ICLR25).

Redefining Contributions: Shapley-Driven Federated Learning.

Nurbek Tastan, Samar Fares, Toluwani Aremu, Samuel Horváth, Karthik Nandakumar. IJCAI, 2024.

CaPriDe Learning: Confidential and Private Decentralized Learning based on Encryption-friendly Distillation Loss.

Nurbek Tastan, Karthik Nandakumar. CVPR, 2023.

SELECTED HONORS & AWARDS

National & Prestigious Awards

Presidential Scholarship, Ministry of Education, Kazakhstan	2020
Award “Altyn Belgi” (Golden Badge), Ministry of Education, Kazakhstan	2017
National Presidential Olympiad, Ministry of Education, Kazakhstan (I place - regional, III place - final)	2016
Republican Scientific Competition (Applied Mathematics) - III place (Bronze)	2016
National Scientific Competition (Computer Science) - II place (Silver)	2020, 2021
Physics Olympiad, II place (Silver)	2015
Mathematics Olympiad, I place (Gold)	2013

ACADEMIC SERVICES

Program Committee (PC) / Reviewer for:

Conferences: AAAI 2024, ICLR 2025, CVPR 2025, ICML 2025, ICCV 2025, NeurIPS 2025, WACV 2026, AAAI 2026, ICLR 2026, CVPR 2026, ICML 2026, ECCV 2026, BMVC 2026, NeurIPS 2026

Journals: IEEE Transactions on Mobile Computing (TMC), IEEE Signal Processing Letters (SPL), IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)

REFERENCES

Dr. Karthik Nandakumar

PhD Supervisor

Associate Professor at MBZUAI & MSU

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Dr. Samuel Horváth

PhD Supervisor

Assistant Professor at MBZUAI

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